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NGA STANDARDIZATION DOCUMENT

SENSOR INDEPENDENT COMPLEX DATA (SICD)

Volume 2

File Format Description Document

Specification of the placement of SICD data products
in the allowed image file formats.

(2025-06-10)

Version 1.5

The National Geospatial-Intelligence Agency's Office of General Counsel has determined that the standards and information set forth in this document are not subject to the export restrictions of the Arms Export Control Act and the International Traffic in Arms Regulations.

NATIONAL CENTER FOR GEOSPATIAL INTELLIGENCE STANDARDS

FOREWORD

The suite of NGA.STND.0024 standardization documents defines the Sensor Independent Complex Data (SICD) format for complex image data products generated by Synthetic Aperture Radar (SAR) systems and their data processing elements.

A SAR complex image is an intermediate data product. The real utility is in the products and measurements that may be derived from it. The quality of the pixel array (resolution, signal-to-noise ratio, etc.), along with the set of metadata provided, are critical in generating the derived products. The "sensor independence" of the SICD product refers to the ability of the allowed pixel array and metadata options to accurately describe the image products from many sensors and data processing systems. Sensor independence does NOT mean that all products have the same format for the pixel array or the same set of metadata parameters.

The SICD documentation has been organized into three volumes and one XML schema. The three volumes are summarized below. The SICD XML schema document defines the XML metadata structure for the XML metadata instance included in a given product.

Volume 1 Design & Implementation Description Document

Contains the description needed by producers of SAR complex image products to design a SICD product and the set of metadata that describe it.

Volume 2 File Format Description Document

Defines the placement of SICD data products in the allowed image file formats. Also provides the description needed by users of SICD products to read and properly extract the SICD data components from a SICD product file.

Volume 3 Image Projections Description Document

Describes the SICD sensor model and the correct projections from image location to ground point and from ground point to image location for all SICD products.

The companion suite of NGA.STND.0025 standardization documents defines the Sensor Independent Derived Data (SIDD) format for image data products that may be derived from a SICD data product. The NGA.STND.0024 and NGA.STND.0025 documents and associated XML artifacts are available in the National System for Geospatial-Intelligence (NSG) Standards Registry (https://nsgreg.nga.mil).

REVISION HISTORY

Date	Version	Description
2011-04-22	1.0	Initial publication.
2014-09-30	1.1	Table 3-2 Updated Fields, Size, Value Range and Types to match MIL-STD-2500C Table 3-3 Updated Value Ranges to match MIL-STD-25000C Table 3-4 Updated Fields to match MIL-STD-2500C Table 3-5 Updated Types to match MIL-STD-2500C
2016-06-30	1.2	Table 3-2 Update to guidance for FTITLE field. Table 3-4 Update to guidance for IID2 field.
2018-12-13	1.2.1	Contacts updated. Table 1-1 Applicable Documents updated. Section 2.1 Introductory text updated. Section 3.2 Image Segment Sizing Calculation text updated. Table 3-4 NITF 2.1 Image Segment Header updated.
2021-11-30	1.3.0	Table 1-1 Applicable Documents updated. Section 2.1 Estimated XML instance size revised. Section 2.2 Reformatted Figure 2-1 and Figure 2-2. Replaced references to MIL-STD-2500C with Joint BIIF Profile (JBP). Table 3-3 MIL-STD-2500C reference replaced with NITF Field Value Registry. 10/7/2022: Document re-released with minor administrative corrections – all equations & symbols now display correctly.
2023-10-26	1.4.0	Foreword revised to identify NGA.STND.0024 and NGA.STND.0025 standards. Section 1-1 Scope: Revised for clarity. Table 1-1: Updated to reflect the correct references, version numbers, and dates. Table 3-3: Name column revised to delete guidance that is specified in the BIIF profile. SICD should defer to the BIIF profile. Added List of Abbreviations and updated the list of Terms and Definitions in Appendix A.
2024-05-01	1.4.0	• No changes to this document. • V1.4.0 XML schema corrected to show APOs are optional.

Date	Version	Description
2025-06-10	1.5	• Version number reduced to two-digit format per direction from the NITFS Technical Board. • Language updated throughout to better indicate testable requirements. • Section 2.2 content removed as redundant with Volume 1. • Table 3-3 contents removed in deference to Joint BIIF Profile. • Table 3-5: Clarified values of DESSHSV, DESSHSD, DESSHTN • Tables 3-2, 3-4, 3-5 reworked to emphasize SICD-specific content
2026-02-28	1.5	• Administrative changes made following the GEOINT Working Group review. • Typographical errors corrected. No change to SICD V1.5 format design and definition. • No changes made to the SICD V1.5 XML schema.

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LIST OF ABBREVIATIONS

Abbreviation & Acronyms	
Refer to NGA.STND.0024-1 SICD Volume 1 and NGA.STND.0024-2 SICD Volume 3	
Abbreviation	**Phrase in Full – Reference for SICD specific description.**
ALVL	Attachment Level
BCS	Basic Character Set
BCS-A	Basic Character Set Alphanumeric
BCS-N	Basic Character Set Numeric
BIIF	Basic Image Interchange Format
CST	Collection Start Time
DES	Data Extension Segment
ECF or ECEF	Earth Centered Fixed (ECF) or Earth Centered Earth Fixed (ECEF) – Used interchangeably to refer to the ECEF frame w/ ECF primarily used in XML parameter names.
HAE	Height Above the Ellipsoid – Measured relative to the WGS 84 reference ellipsoid.
ICP	Image Corner Point
IS	Image Segment
ISCP	Image Segment Corner Point
JBP	Joint BIIF Profile
LLH	Latitude, Longitude, Height Above the Ellipsoid
MSB	Most Significant Byte
NITFS	National Imagery Transmission Format Standard
NTB	NITFS Technical Board
SAR	Synthetic Aperture Radar
SCP	Scene Center Point – See Volume 1: Section 4.2
SICD	Sensor Independent Complex Data
SIDD	Sensor Independent Derived Data
TRE	Tagged Record Extension
UTC	Coordinated Universal Time
WGS 84	World Geodetic System 1984
XML	Extensible Markup Language

1 Introduction

1.1 Scope

The Sensor Independent Complex Data (SICD) File Format Description Document provides detailed specifications for how SICD products are placed in the allowed SICD container files. A SICD product is divided into a pair of data components. The data components include the complex image pixel data and the metadata that describe it. See Section 2.

This document defines how the SICD data components are mapped to the components of the allowed container files. A typical container file format includes required file components (e.g., a file header) as well as optional and conditional file components. Guidance is provided on how the required and optional (if any) file components are to be populated for a SICD product. Provided is the description needed by the producers of SICD products to properly place the SICD data components into the product file. Also provided is the description needed by users of SICD products to read and properly extract the SICD data components from a SICD product file.

Currently, only a single container file format has been selected for SICD product files. The format is the National Imagery Transmission Format, Version 2.1 (referred to as NITF 2.1). The NITF 2.1 standard is described in Joint Basic Image Interchange Format Profile (JBP). See Table 1-1.

1.2 Applicable Documents

The documents listed below are referenced in this document. Users of this document should investigate recent editions and change notices of the documents listed below.

Table 1-1 Government Documents and Publications	
Number	**Title & Website**
ISO/IEC Joint BIIF Profile JBP-2025.1	ISO/IEC Joint BIIF Profile (JBP) 2025.1, 10 June 2025 https://nsgreg.nga.mil/ntb.jsp
STDI-0002 v2025.2	The Compendium of Controlled Support Data Extensions (SDE) for the National Imagery Transmission Format (NITF) Version 2025.2; 10 June 2025 https://nsgreg.nga.mil/ntb.jsp
NGA.STND.0024-1_1.5	Sensor Independent Complex Data (SICD), Volume 1, Design & Implementation Description Document, Version 1.5, 2025-06-10 https://nsgreg.nga.mil/ntb.jsp
USAF.RDUCE-001 SARzip	Synthetic Aperture Radar (SAR) Compression (SARzip), v1.0.0, 2017-12-11 https://nsgreg.nga.mil/doc/view?i=4506

2 SICD Data Components

2.1 SICD Product Data Components

A SICD product consists of a pair of data components. The first component is a block of complex image pixel data. The second component is a block of metadata expressed using the eXtensible Markup Language (XML). The metadata contains the selected set of parameters that describe the imaging collection and the data processing that formed the image. The SICD data components are described in detail in the Sensor Independent Complex Data Design & Implementation Description Document. The SICD components are briefly described below.

The image pixel data in a given product may be stored in either standard format (i.e., uncompressed format) or compressed format. For all products, the XML metadata specifies the size and format of the pixel data in standard format. The following guidance and descriptions are for the image pixel data in standard format.

The SICD image pixel data block is a single two-dimensional array of complex numbers. The image pixel array contains NumRows rows by NumCols columns. The SICD image pixel data is stored in one of three formats specified by metadata parameter PixelType: RE32F_IM32F, RE16I_IM16I, or AMP8I_PHS8I. Where the SICD PixelType has real and imaginary components, the real and imaginary components shall be interleaved such that they are stored in adjacent bytes with the real component stored first. Where the SICD PixelType has amplitude and phase components, the amplitude and phase components shall be interleaved such that they are stored in adjacent bytes with the amplitude component stored first. NITF files are big-endian.

The adjacent pixels within a row shall be stored in adjacent bytes within the file. The number of bytes per pixel shall be denoted by parameter BytesPerPixel (where BytesPerPixel = 2, 4 or 8). The NumCols pixels of the first row in the image array shall be stored in the first bytes in the file (BytesPerPixel x NumCols bytes). The total size of the image array is BytesPerPixel x NumCols x NumRows bytes.

The SICD XML metadata block contains the set of metadata selected to support the image. Contained within the SICD XML metadata block are the parameters used to place the SICD data components into the file components of the selected SICD container file. This set of parameters is referred to as the Container Placement Parameters. These parameters are used to populate the container file header and sub-headers and to place the data components into the file components. The Container Placement Parameters are listed in Table 2-1 below. All Container Placement Parameters are required parameters.

2.2 Image Arrays: Full Image & Sub-Image

Content removed in v1.5. See NGA.STND.0024-1_1.5 section 1.2.2.

2.3 SICD Container Placement Parameters

Table 2-1 SICD Container Placement Parameters

Parameter	XML Field Name	Req / Opt	Type	Description
CoreName	SICD.CollectionInfo.CoreName	R	TXT	Collection & imaging data set identifier per Program Specific Implementation Document.
CollectorName	SICD.CollectionInfo.CollectorName	R	TXT	Radar platform identifier. For Bistatic collections, list the Receive platform.
CollectStart	SICD.Timeline.CollectStart	R	XDT	Collection date and start time (UTC).
Classification	SICD.CollectionInfo.Classification	R	TXT	Contains the human-readable banner. Contains classification, file control & handling, file releasing, and/or proprietary markings. Specified per Program Specific Implementation Document. Default value: "UNCLASSIFIED".
PixelType	SICD.ImageData.PixelType	R	ENU	Indicates the type and binary format used for individual pixels. PixelType = RE32F_IM32F ⇔ Real & Imaginary stored in 32 bit binary32 floating point format. PixelType = RE16I_IM16I⇔ Real & Imaginary stored in 16 bit signed integer 2's complement format. PixelType = AMP8I_PHS8I⇔ Amplitude & Phase stored in 8 bit unsigned integer format.
NumRows	SICD.ImageData.NumRows	R	INT	Number of rows in the product. Rows within the product indexed row_p = 0, 1, . . . , NumRows – 1.
NumCols	SICD.ImageData.NumCols	R	INT	Number of columns in the product. Columns within the product indexed col_p = 0, 1, . . . , NumCols – 1.
ICPLat(1)	SICD.Geodata.ImageCorners.ICP.Lat	R	DBL	Image Corner Point 1 latitude (decimal degrees). ICP = 1 ⇔ First row, first column of the product.
ICPLon(1)	SICD.Geodata.ImageCorners.ICP.Lon	R	DBL	Image Corner Point 1 longitude (decimal degrees).

Table 2-1 SICD Container Placement Parameters

Parameter	XML Field Name	Req / Opt	Type	Description
ICPLat(2)	SICD.Geodata.ImageCorners.ICP.Lat	R	DBL	Image Corner Point 2 latitude (decimal degrees). ICP = 2 ⇔ First row, last column of the product.
ICPLon(2)	SICD.Geodata.ImageCorners.ICP.Lon	R	DBL	Image Corner Point 2 longitude (decimal degrees).
ICPLat(3)	SICD.Geodata.ImageCorners.ICP.Lat	R	DBL	Image Corner Point 3 latitude (decimal degrees). ICP = 3 ⇔ Last row, last column of the product.
ICPLon(3)	SICD.Geodata.ImageCorners.ICP.Lon	R	DBL	Image Corner Point 3 longitude (decimal degrees).
ICPLat(4)	SICD.Geodata.ImageCorners.ICP.Lat	R	DBL	Image Corner Point 4 latitude (decimal degrees). ICP = 4 ⇔ Last row, first column of the product.
ICPLon(4)	SICD.Geodata.ImageCorners.ICP.Lon	R	DBL	Image Corner Point 4 longitude (decimal degrees).
Note: For a description of allowed Req/Opt values and defined XML Types, see NGA.STND.0024-1, Sect 2.1.2 and Sect 2.1.3.				

3 SICD Products in NITF 2.1 Format

This Section provides a detailed definition of how the SICD product is stored in a NITF 2.1 container file.

3.1 SICD in NITF 2.1 Format

The structure of a SICD product in a NITF 2.1 container file is shown in Table 3-1. The NITF 2.1 file consists of a File Header record, one or more Image Segment records and at least one Data Extension Segment record. The data placed in the File Header is defined in Table 3-2 and is based upon Table 5.11-1 of JBP. The data placed in the Image Sub-Header is defined in Table 3-4 and is based upon Table 5.13-1 of JBP. The data placed in the Data Extension Segment Sub-Header is defined in Table 3-5 and is based upon Table F-1 of STDI-0002, Volume 2 Appendix F: XML_DATA_CONTENT DES.

The mapping of SICD data components to NITF 2.1 file components is shown in Figure 3-1. The example shows a product file with the image pixel array stored in two Image Segments. The Container Placement parameters (contained in the XML data) are used to populate fields in the File Header, the Image Sub-Headers and the DES Sub-Header.

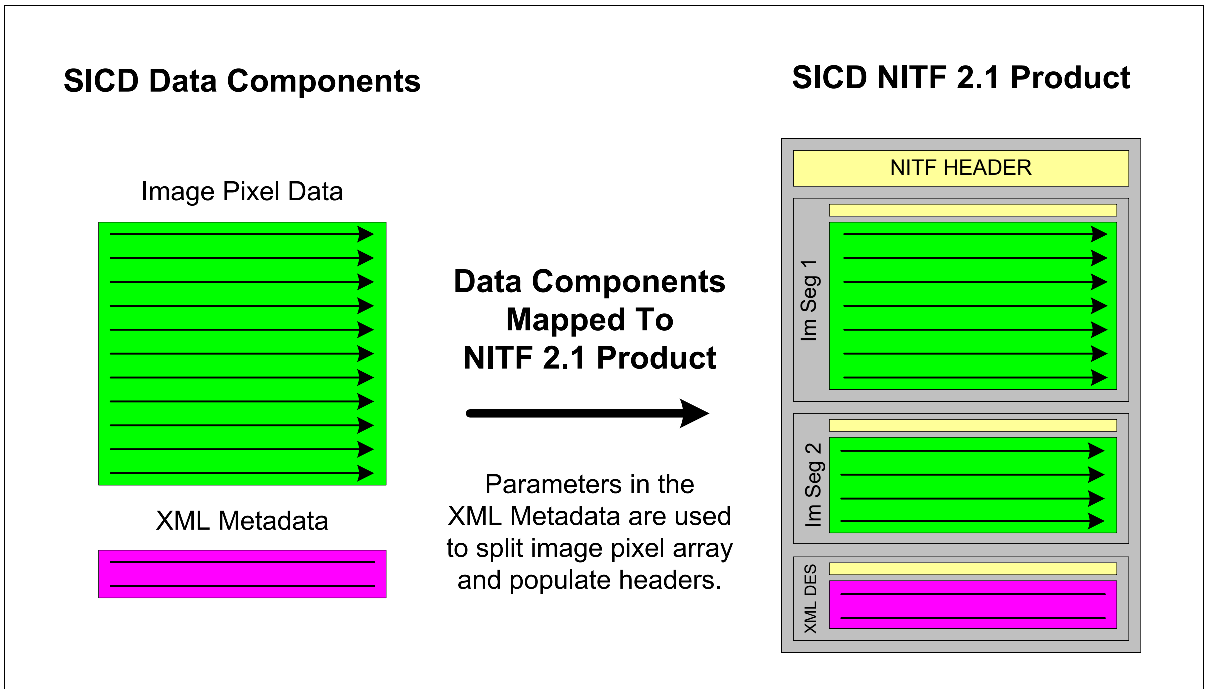

Figure 3-1 SICD NITF 2.1 Product

SICD data components mapped to NITF 2.1 file components.

3.1.1 Format Specifications

The NITF File Header includes fields to identify the product and its source. The File Header includes the number of Image Segments contained in the file. The File Header also indicates the number of Data Extension Segments (DESs) contained in the file. The SICD image pixel data block is stored in one or more Image Segments. The SICD XML Metadata block shall be stored in the first DES. Security fields are included in the File Header, all Image Sub-Headers and the DES Sub-Header. The security fields are defined in Table 3-3. The NITF encoding is being used to address the Intelligence Community information security marking data elements, which are the essential security markings applicable to the data contained within the file.

The implementation of SICD products for a particular SAR sensor and data processing system includes many program specific design choices. See Section 4.1 of the NGA.STND.0024-1, SICD Design & Implementation Description Document. Program specific guidance is included in what is referred to as a Program Specific Implementation Document. The program specific guidance may include the range of allowed values for the parameters listed in Tables 3-2 through 3-5.

The number of Image Segments used to hold the complex data shall be determined from the size of the complex pixel data. Each Image Segment can hold up to 9,999,999,998 bytes of data ($10^{10} - 2$ bytes). If the size of the complex data exceeds the 9,999,999,998 byte limit, it shall be placed in multiple Image Segments. If two or more Image Segments are used, there is also a maximum number of rows per image segment restriction of 99,999. The number of ISs for the complex data shall be determined as described in Section 3.2. The complex pixel data shall be broken across rows when stored in multiple ISs. The next row shall begin in the next IS so there is continuous flow of data across the segments. The complex data shall start in the first Image Segment. The Image Sub-Header fields indicate the portion of the image pixel array stored and its relative placement in the complete image pixel array.

The first DES shall be of type XML_DATA_CONTENT and contain the SICD XML metadata. The SICD XML metadata is described in Volume 1, Design & Implementation Description Document. The placement of the XML block in the DES is independent of the precise metadata in the XML block.

The SICD data product does not make explicit use of NITF Tagged Record Extensions (TREs) to describe the format or attributes of the image pixel array. All SICD specific metadata is contained in the SICD metadata block. While the use and inclusion of TREs in the File Header and/or Image Sub-Header is not prohibited, use of TREs with SICD should be coordinated with the NITFS Technical Board (NTB). Use of TREs should be identified in the Program Specific Implementation Document. Producers of SICD products that choose to include authorized TREs will be responsible for correct formatting and placement. Reader applications developed for SICD products should be designed to accommodate NTB-authorized TREs.

3.2 Image Segment Sizing Calculations

The computations to determine the number of Image Segments used to contain the image pixel data are described below. The computations also show how to fill NITF fields that vary across Image Segments. Parameter NumIS is the number of Image Segments. Image Segments are indexed n = 1 to NumIS. Parameter NumRowsIS(n) is the number of rows in the nth Image Segment. Parameter FirstRowIS(n) is the product row index (row_p) of the first row in the nth Image Segment. RowOffsetIS(n) is a row offset parameter required for the Image Sub-Header of the nth Image Segment. When the complex pixel array is split across multiple Image Segments, the following NITF Image Sub-Header fields are affected: IID1, NROWS/NCOLS, IGEOLO, NPPBH/NPPBV, IDLVL/IALVL and ILOC.

The Image Segment sizing calculations specified in the following sections apply when the image pixel data is stored in standard format (IC=NC or IC=NM). If the image pixel data is stored in compressed format (IC=C7 or IC=M7), see the SARzip standard (USAF.RDUCE-001) for guidance.

3.2.1 Image Segment Parameters and Equations

SICD Container Placement Parameters – See Table 2-1.

PixelType	Indicates the pixel type and binary format.
NumRows	Rows indexed $row_p = 0, 1, 2, \ldots, NumRows - 1$.
NumCols	Columns indexed $col_p = 0, 1, 2, \ldots, NumCols - 1$.
ICPLat (1), ICPLon(1)	Coordinate for pixel (0,0)
ICPLat (2), ICPLon(2)	Coordinate for pixel (0,NumCols-1)
ICPLat (3), ICPLon(3)	Coordinate for pixel (NumRows-1, NumCols-1)
ICPLat (4), ICPLon(4)	Coordinate for pixel (NumRows-1, 0)

Constants

ISsizeMax = 9,999,999,998

ILOCMax = 99,999

Number of Image Segments & Image Segment Size Parameters

$$\text{BytesPerPixel} = \begin{cases} 8, & \text{if PixelType} = \text{RE32F\_IM32F} \\ 4, & \text{if PixelType} = \text{RE16I\_IM16I} \\ 2, & \text{if PixelType} = \text{AMP8I\_PHS8I} \end{cases}$$

$$\text{BytesPerRow} = \text{BytesPerPixel} \bullet \text{NumCols}$$

$$\text{ProductSize} = \text{BytesPerPixel} \bullet \text{NumRows} \bullet \text{NumCols}$$

$$\text{Limit1} = \text{floor}\left(\frac{\text{ISsizeMax}}{\text{BytesPerRow}}\right) \quad , \qquad \text{Limit2} = \text{ILOCMax}$$

$$\text{NumRowsLimit} = \min(\text{ Limit1, Limit2 })$$

If $(\text{ProductSize} \leq \text{ISsizeMax})$ ***then,***

$\text{NumIS} = 1$

$\text{NumRowsIS}(1) = \text{NumRows}$

$\text{FirstRowIS}(1) = 0$

$\text{RowOffsetIS}(1) = 0$

Else if $(\text{ProductSize} > \text{ISsizeMax})$ ***then,***

$$\text{NumIS} = \text{ceiling}\left(\frac{\text{NumRows}}{\text{NumRowsLimit}}\right)$$

$\text{FirstRowIS}(1) = 0$

$\text{RowOffsetIS}(1) = 0$

For $(n = 1, \ldots, \text{NumIS} - 1)$

$\text{NumRowsIS}(n) = \text{NumRowsLimit}$

$\text{FirstRowIS}(n+1) = n \bullet \text{NumRowsLimit}$

$\text{RowOffsetIS}(n+1) = \text{NumRowsLimit}$

End

$\text{NumRowsIS}(\text{NumIS}) = \text{NumRows} - (\text{NumIS} - 1) \bullet \text{NumRowsLimit}$

End if

Image Segment Corner Coordinate Parameters

The corner coordinates for the portion of the image stored in each Image Segment are computed as follows. For Image Segment n, the corners are indexed 1 to 4 as follows.

ISCPLat(n,1), ISCPLon(n,1)	Coordinate for pixel (FirstRowIS(n),0)
ISCPLat(n,2), ISCPLon(n,2)	Coordinate for pixel (FirstRowIS(n),NumCols-1)
ISCPLat(n,3), ISCPLon(n,3)	Coordinate for pixel (LastRowIS(n),NumCols-1)
ISCPLat(n,4), ISCPLon(n,4)	Coordinate for pixel (LastRowIS(n),0)

where $\text{LastRowIS}(n) = \text{FirstRowIS}(n) + \text{NumRowsIS}(n) - 1$.

For an image that fits in one Image Segment, assign the corner coordinates of the Image Segment to the corner coordinates of the image pixel array.

ISCPLat(1,1) = ICPLat(1)	ISCPLon(1,1) = ICPLon(1)
ISCPLat(1,2) = ICPLat(2)	ISCPLon(1,2) = ICPLon(2)

$$\text{ISCPLat}(1,3) = \text{ICPLat}(3) \qquad \text{ISCPLon}(1,3) = \text{ICPLon}(3)$$

$$\text{ISCPLat}(1,4) = \text{ICPLat}(4) \qquad \text{ISCPLon}(1,4) = \text{ICPLon}(4)$$

For an image that is split across Image Segments, the corner coordinates for each Image Segment are computed as follows. Convert the image pixel array corner positions from latitude and longitude to Earth Centered Fixed (ECF) coordinates. Use a Height Above Ellipsoid = 0.

For each icp = 1, 2, 3, 4:

$$\text{Convert LLH to ECF: } \begin{bmatrix} \text{ICPLat(icp)} \\ \text{ICPLon(icp)} \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} \text{ICP\_ECFX(icp)} \\ \text{ICP\_ECFY(icp)} \\ \text{ICP\_ECFZ(icp)} \end{bmatrix}$$

For each Image Segment, compute the ECF coordinates for the corners of the first row in the Image Segment (ISCP(n,1) and ISCP(n,2)). Linearly interpolate the ECF positions. Convert the interpolated ECF positions back to LLH.

For n = 1, . . . , NumIS:

$$\text{wgt1(n)} = \frac{\text{NumRows} - 1 - \text{FirstRowIS(n)}}{\text{NumRows} - 1} \qquad \text{wgt2(n)} = \frac{\text{FirstRowIS(n)}}{\text{NumRows} - 1}$$

$$\begin{bmatrix} \text{ISCP\_ECFX(n,1)} \\ \text{ISCP\_ECFY(n,1)} \\ \text{ISCP\_ECFZ(n,1)} \end{bmatrix} = \text{wgt1(n)} \bullet \begin{bmatrix} \text{ICP\_ECFX(1)} \\ \text{ICP\_ECFY(1)} \\ \text{ICP\_ECFZ(1)} \end{bmatrix} + \text{wgt2(n)} \bullet \begin{bmatrix} \text{ICP\_ECFX(4)} \\ \text{ICP\_ECFY(4)} \\ \text{ICP\_ECFZ(4)} \end{bmatrix}$$

$$\text{Convert ECF to LLH: } \begin{bmatrix} \text{ISCP\_ECFX(n,1)} \\ \text{ISCP\_ECFY(n,1)} \\ \text{ISCP\_ECFZ(n,1)} \end{bmatrix} \rightarrow \begin{bmatrix} \text{ISCPLat(n,1)} \\ \text{ISCPLon(n,1)} \\ \text{ISCPHAE(n,1)} \end{bmatrix}$$

$$\begin{bmatrix} \text{ISCP\_ECFX(n,2)} \\ \text{ISCP\_ECFY(n,2)} \\ \text{ISCP\_ECFZ(n,2)} \end{bmatrix} = \text{wgt1(n)} \bullet \begin{bmatrix} \text{ICP\_ECFX(2)} \\ \text{ICP\_ECFY(2)} \\ \text{ICP\_ECFZ(2)} \end{bmatrix} + \text{wgt2(n)} \bullet \begin{bmatrix} \text{ICP\_ECFX(3)} \\ \text{ICP\_ECFY(3)} \\ \text{ICP\_ECFZ(3)} \end{bmatrix}$$

$$\text{Convert ECF to LLH: } \begin{bmatrix} \text{ISCP\_ECFX(n,2)} \\ \text{ISCP\_ECFY(n,2)} \\ \text{ISCP\_ECFZ(n,2)} \end{bmatrix} \rightarrow \begin{bmatrix} \text{ISCPLat(n,2)} \\ \text{ISCPLon(n,2)} \\ \text{ISCPHAE(n,2)} \end{bmatrix}$$

Assign corner coordinates for the last row of each Image Segment as follows. For the first NumIS - 1 image segments, assign the coordinates of the last row to the coordinates of the first row of the next Image Segment. For image segment NumIS, assign the coordinates of the last row to the coordinates of the last row of the image.

For n = 1, . . . , NumIS - 1:

ISCP _Lat(n,3) = ISCP Lat(n+1,2) ISCP _Lon(n,3) = ISCP Lon(n+1,2)
ISCP _Lat(n,4) = ISCP Lat(n+1,1) ISCP _Lon(n,4) = ISCP Lon(n+1,1)

For n = NumIS:

ISCP Lat(n,3) = ICPLat(3) ISCPLon(n,3) = ICPLon(3)
ISCPLat(n,4) = ICPLat(4) ISCPLon(n,4) = ICPLon(4)

3.2.2 File Header and Image Sub-Header Parameters

Using the parameters computed above, the following File Header and Image Sub-Header parameters shall be assigned as described below. The parameter ILOC (RRRRRCCCCC) shall be filled by concatenating the RowOffsetIS value (RRRRR) and 00000 (CCCCC). The image display level (IDLVL) and the attachment level (IALVL) are set to indicate the connection across Image Segments.

NITF File Header Parameters

File Header NUMI = NumIS

NITF Image Sub-Header Parameters

For n = 1, . . . , NumIS - 1:

Image Sub-Header (n) NROWS = NumRowsIS(n)

Image Sub-Header (n) ILOCRow = RowOffsetIS(n)

Image Sub-Header (n) IDLVL = n

Image Sub-Header (n) IALVL = n – 1

$$\text{Image Sub-Header (n) IGEOLO} = \text{CONCAT}\begin{Bmatrix} \text{ISCPLat(n,1)}, & \text{ISCPLon(n,1)} \\ \text{ISCPLat(n,2)}, & \text{ISCPLon(n,2)} \\ \text{ISCPLat(n,3)}, & \text{ISCPLon(n,3)} \\ \text{ISCPLat(n,4)}, & \text{ISCPLon(n,4)} \end{Bmatrix}$$

Note: See IGEOLO description in Table 3-4.

3.2.3 Example Image Segment Sizing Computations

Example 1. PixelType = RE32F_IM32F NumRows = 2,500 NumCols = 5,000

This image pixel array is placed in one Image Segment.

BytesPerPixel = 8

BytesPerRow = 8 x 5000 = 40,000

ProductSize = 8 x 2,500 x 5,000 = 100,000,000

Limit1 = floor(9,999,999,998 / 40,000) = floor(249,999.999) = 249,999

NumRowsLimit = min(249,999 , 99,999) = 99,999

(ProductSize ≤ ISsizeMax) ➔

NumIS = 1

NumRowsIS(1) = 2,500

FirstRowIS(1) = 0

RowOffsetIS(1) = 0

File Header: NUMI = 1

Image Sub-Header 1: NROWS = 2500, ILOCRow = 0, IDLVL = 1, IALVL = 0

Example 2. PixelType = RE32F_IM32F NumRows = 30,000 NumCols = 90,000

This image pixel array is placed in 3 Image Segments. Image Segments 1 and 2 contain approximately 10^{10} bytes each.

BytesPerPixel = 8

BytesPerRow = 8 x 90,000 = 720,000

ProductSize = 8 x 30,000 x 90,000 = 21,600,000,000

Limit1 = floor(9,999,999,998 / 720,000) = floor(13888.888) = 13,888

NumRowsLimit = min(13,888 , 99,999) = 13,888

(ProductSize > ISsizeMax) ➔

NumIS = ceiling(30,000 / 13,888) = ceiling(2.160) = 3

FirstRowIS(1) = 0

RowOffsetIS(1) = 0

For n = 1:2

NumRowsIS(1) = 13,888

FirstRowIS(2) = 1 x 13,888 = 13,888

RowOffsetIS(2) = 13,888

NumRowsIS(2) = 13,888

FirstRowIS(3) = 2 x 13,888 = 27776

RowOffsetIS(3) = 13,888

For n = 3

NumRowsIS(3) = 30,000 – (2 x 13,888) = 2224

File Header: NUMI = 3

Image Sub-Header 1: NROWS = 13,888, ILOCRow = 0, IDLVL = 1, IALVL = 0

Image Sub-Header 2: NROWS = 13,888, ILOCRow = 13,888, IDLVL = 2, IALVL = 1

Image Sub-Header 3: NROWS = 2224, ILOCRow = 13,888, IDLVL = 3, IALVL = 2

Example 3. PixelType = RE16I_IM16I NumRows = 150,000 NumCols = 20,000

This image pixel array is placed in 2 Image Segments. Image Segment 1 contains 99,999 rows (the maximum number of rows per Image Segment when data is split across Image Segments).

BytesPerPixel = 4

BytesPerRow = 4 x 20,000 = 80,000

ProductSize = 4 x 150,000 x 20,000 = 12,000,000,000

Limit1 = floor(9,999,999,998 / 80,000) = floor(124,999.99) = 124,999

NumRowsLimit = min(124,999 , 99,999) = 99,999

(ProductSize > ISsizeMax) ➔

NumIS = ceiling(150,000 / 99,999) = ceiling(1.500) = 2

FirstRowIS(1) = 0

RowOffsetIS(1) = 0

For n = 1:1

NumRowsIS(1) = 99,999

FirstRowIS(2) = 1 x 99,999 = 99,999

RowOffsetIS(2) = 99,999

For n = 2

NumRowsIS(2) = 150,000 – (1 x 99,999) = 50001

File Header: NUMI = 2

Image Sub-Header 1: NROWS = 99,999, ILOCRow = 0, IDLVL = 1, IALVL = 0

Image Sub-Header 2: NROWS = 50001, ILOCRow = 99,999, IDLVL = 2, IALVL = 1

3.3 NITF Header Parameters

The SICD NITF shall have headers and sub-headers populated according to the tables in this section. Shown in Table 3-1 is an example NITF file structure for a SICD product that is composed of only the required components. The required components include a file header, one or more image segments, and a single data extension segment (DES). Each image segment includes an image sub-header and image data. Each DES includes a DES sub-header and DES data.

Table 3-1 Example NITF 2.1 File Structure for a SICD Product

NITF File Header	Image (1) Sub-Header	Image (1) Data	. . .	Image (n) Sub-Header	Image (n) Data	DES (1) Sub-Header	DES (1) Data

In the tables that follow, for a "Value Range" that is not specified, see either the ISO/IEC Joint BIIF Profile (JBP) or the applicable Program Specific Implementation Document for any implementation value restrictions.

Table 3-2 NITF 2.1 File Header

Short Name	Field Name	Size (bytes)	Value Range	Type	Comment
FHDR	File Profile Name	4	"NITF"	R	
FVER	File Version	5	"02.10"	R	
CLEVEL	Complexity Level	2		R	
STYPE	Standard Type	4		R	
OSTAID	Originating Station ID	10		R	
FDT	File Date and Time	14		R	
FTITLE	File Title	80		<R>	
File Header Security Fields		167	As defined in Table 3-3	R	
FSCOP	File Copy Number	5	"00000"	R	No tracking of numbered file copies
FSCPYS	File Number of Copies	5	"00000"	R	No tracking of numbered file copies
ENCRYP	Encryption	1		R	
FBKGC	File Background Color	3	(0x00, 0x00, 0x00)	R	Black
ONAME	Originator's Name	24		<R>	
OPHONE	Originator's Phone	18		<R>	
FL	File Length	12		R	
HL	NITF File Header Length	6		R	
NUMI	Number of Image Segments	3	"001" to "999"	R	At least the number computed for the image data **See Section 3.2**

Table 3-2 NITF 2.1 File Header					
Short Name	**Field Name**	**Size (bytes)**	**Value Range**	**Type**	**Comment**
. . . . . Start for each IS LISHn, LIn.					
NOTE: LISHn and LIn fields repeat in pairs such that LISH001, LI001; LISH002, LI002; .....LISHn,LIn.					
LISHn	Length of Image Sub-Header	6		R	
LIn	Length of nth Image Segment	10		R	NumRowsIS(n) x NumCols x BytesPerPixel **See Section 3.2**
. . . . End for each IS LISHn, LIn; the number of loop repetitions is the value specified in the NUMI field.					
NUMS	Number of graphic Segments	3	“000”	R	No Graphic Segments
NUMX	Reserved	3		R	
NUMT	Number of Text Segments	3		R	
LTSHn and LTn repeat as pairs as many times as specified in the NUMT field					
LTSHn	Length of nth Text Segment Sub-Header	4		C	
LTn	Length of Text Segment	5		C	
. . . . End for each IS LTSHn, LTn; the number of loop repetitions is the value specified in the NUMT field.					
NUMDES	Number of data Extension Segments	3	“001” to “999”	R	
LDSHn and LDn repeat as pairs as many times as specified in the NUMDES field					
LDSHn	Length of nth Data Extension Segment Sub-Header	4		R	
LDn	Length of Data Extension Segment	9		R	
End of LDSH and LDn field repetition					
NUMRES	Number of Reserved	3	“000”	R	No Reserved Extension Segments
UDHDL	User-Defined Header Data Length	5		R	
UDHOFL	User-Defined Header Overflow	3		C	
UDHD	User-Defined Header Data	Equal to UDHDL minus 3		C	
XHDL	Extended Header Data Length	5		R	
XHDLOFL	Extended Header Data Overflow	3		C	

Table 3-2 NITF 2.1 File Header					
Short Name	**Field Name**	**Size (bytes)**	**Value Range**	**Type**	**Comment**
XHD	Extended Header Data	Equal to XHDL minus 3		C	
Type – R = Required, C = Conditional, <> surrounding the type = BCS SPACES ALLOWED FOR ENTIRE FIELD					

Table 3-3 NITF 2.1 Security Tags
Table contents removed in v1.5. See ISO/IEC Joint BIIF Profile, Section 5.10

Table 3-4 NITF 2.1 Image Sub-Header

Short Name	Field Name	Size (bytes)	Value Range	Type	Comment
IM	File Part Type	2		R	
IID1	Image Identifier 1	10	"SICDnnn" nnn has a range 000, 001-999	R	For complex images contained in a single IM segment, nnn = 000 For complex images contained in two or more IM segments, populate nnn with 001 for the first segment and progress sequentially for each segment used to contain the image.
IDATIM	Image Date and Time	14		R	SICD Parameter: CollectStart
TGTID	Target Identifier	17		<R>	
IID2	Image Identifier 2	80		<R>	
Security Fields Image		167	As defined in Table 3-3	R	
ENCRYP	Encryption	1		R	
ISORCE	Image Source	42		R	
NROWS	Number of Significant Rows in Image	8		R	NumRowsIS(n) **See Section 3.2**
NCOLS	Number of Significant Columns in Image	8		R	SICD Parameter: NumCols **See Section 3.2**
PVTYPE	Pixel Value Type	3	"R", "SI", or "INT"	R	SICD Parameter: PixelType RE32F_IM32F = R RE16I_IM16I = SI AMP8I_PHS8I = INT
IREP	Image Representation	8	"NODISPLY"	R	SICD is complex image data that is not display ready without further processing
ICAT	Image Category	8	"SAR"	R	Synthetic Aperture Radar
ABPP	Actual Bits-Per-Pixel Per Band	2	"08", "16", or "32"	R	SICD Parameter: PixelType RE32F_IM32F = 32 RE16I_IM16I = 16 AMP8I_PHS8I = 08 For SICD, ABPP = NBPP **See Section 2.3**
PJUST	Pixel Justification	1	"R"	R	
ICORDS	Image Coordinate Representation	1	"G"	R	G for geographic

Table 3-4 NITF 2.1 Image Sub-Header					
Short Name	**Field Name**	**Size (bytes)**	**Value Range**	**Type**	**Comment**
IGEOLO	Image Geographic Location	60	ddmmssXdddmmssY ddmmssXdddmmssY ddmmssXdddmmssY ddmmssXdddmmssY	C	Calculate from complex image corner coordinates. **See Section 3.2** dd/d = degrees mm = minutes ss = seconds X = “N” or “S” Y = “E” or “W”
NICOM	Number of Image Comments	1		R	
Start for each Image Comment ICOMn field (if the value of NICOM is not equal to zero).					
ICOMn	Image Comment n	80		C	
End for each ICOMn field. The number of loop repetitions is the value specified in the NICOM field.					
IC	Image Compression	2	“NC”, “NM”, “C7”, or “M7”	R	“NC” → No Compression “NM” → No Compression with masking “C7” → SARZIP compressed, no masking. “M7” → SARZIP compressed with masking
NBANDS	Number of Bands	1	“2”	R	2 bands for the Real & Imaginary or Magnitude & Phase image data
IREPBANDn	nth Band Representation	2	BCS spaces (0x20)	<R>	Data not intended for display
ISUBCATn	nth Band Subcategory	6	“I”, “Q”, “M”, or “P”	R	SICD Parameter: PixelType Real (RE) → I Imaginary (IM) → Q Amplitude (AMP) → M Phase (PHS) → P
IFCn	nth Band Image Filter Condition	1		R	
IMFLTn	nth Band Standard Image Filter Code	3		<R>	
NLUTSn	Number of LUTS for the nth Image Band	1	“0”	R	No Look Up Tables
ISYNC	Image Sync code	1		R	
IMODE	Image Mode	1	“P”	R	Band interleaved by Pixel

Table 3-4 NITF 2.1 Image Sub-Header					
Short Name	**Field Name**	**Size (bytes)**	**Value Range**	**Type**	**Comment**
NBPR	Number of Blocks Per Row	4	“0001”	R	
NBPC	Number of Blocks Per Column	4	“0001”	R	
NPPBH	Number of Pixels Per Block Horizontal	4		R	if NCOLS > 8192 value = 0000 if NCOLS <= 8192 value = NCOLS
NPPBV	Number of Pixels Per Block Vertical	4		R	if NROWS > 8192 value = 0000 if NROWS <= 8192 value = NROWS
NBPP	Number of Bits Per Pixel Per Band	2	“08”, “16”, or “32”	R	SICD Parameter PixelType RE32F_IM32F = 32 RE16I_IM16I = 16 AMP8I_PHS8I = 8 **See Section 2.3**
IDLVL	Image Display Level	3		R	Increase by image segment, start at 001 **See Section 3.2**
IALVL	Attachment Level	3		R	Increase by image segment, start at 000 Value corresponds to display level -1. If multiple segments are needed, the attachment level should always match the previous image segments display level. **See Section 3.2**
ILOC	Image Location	10	RRRRR00000	R	RRRRRCCCCC R is for Row C is for Column The first segment RRRRR = 00000. If multiple segments are needed the RRRRR value should match the NROWS in the previous segment. **See Section 3.2**
IMAG	Image Magnification	4	“1.0”	R	
UDIDL	User Defined Image Data Length	5		R	
UDOFL	User Defined Overflow	3		C	

Table 3-4 NITF 2.1 Image Sub-Header					
Short Name	**Field Name**	**Size (bytes)**	**Value Range**	**Type**	**Comment**
UDID	User Defined Image Data	Equal to UDIDL minus 3		C	
IXSHDL	Image Extended Sub-Header Data Length	5		R	
IXSOFL	Image Extended Subheader Overflow	3		C	
IXSHD	Image Extended Subheader Data	Equal to IXSHDL minus 3		C	
Type – R = Required, C = Conditional, <> surrounding the type = BCS SPACES ALLOWED FOR ENTIRE FIELD					

Table 3-5 NITF 2.1 XML_DATA_CONTENT Data Extension Segment Sub-Header					
Short Name	**Field Name**	**Size**	**Value Range**	**Type**	**Comment**
DE	File Part Type	2		R	
DESID	Unique DES Type Identifier	25		R	
DESVER	Version of the Data Definition	2		R	
Security Fields DES		167	As defined in Table 3-3	R	
DESSHL	DES User-defined Subheader Length	4	“0773”	R	All user-defined sub-header fields shall be included.
Note: The ‘conditional’ field, DESSHF (DES User-defined Subheader Field) shall be included in all SICD products. The DESSHF is comprised of the sub-fields DESCRC through DESSHABS specified below.					
DESCRC	Cyclic Redundancy Check	5	“99999”	R	CRC shall not be computed for SICD products.
DESSHFT	XML File Type	8	“XML”	R	
DESSHDT	Date and Time	20	YYYY-MM-DDThh:mm:ssZ	R	
DESSHRP	Responsible Party – Organization Identifier	40		R	
DESSHSI	Specification Identifier	60	"SICD Volume 1 Design & Implementation Description Document"	R	

Table 3-5 NITF 2.1 XML_DATA_CONTENT Data Extension Segment Sub-Header

Short Name	Field Name	Size	Value Range	Type	Comment
DESSHSV	Specification Version	10	"1.5"	R	
DESSHSD	Specification Date	20	"2025-06-10"	R	
DESSHTN	Target Namespace	120	"urn:SICD:1.5"	<R>	
DESSHLPG	Location – Polygon	125	±dd.dddddddd±ddd.ddddddd ±dd.dddddddd±ddd.ddddddd ±dd.dddddddd±ddd.ddddddd ±dd.dddddddd±ddd.ddddddd ±dd.dddddddd±ddd.ddddddd	<R>	Use image corner point locations. See Table 2-1. Order is: ICP1, ICP2, ICP3, ICP4, ICP1.
DESSHLPT	Location – Point	25		<R>	
DESSHLI	Location – Identifier	20		<R>	
DESSHLIN	Location Identifier Namespace URI	120		<R>	
DESSHABS	Abstract	200		<R>	
DESDATA	DES User-defined Data	Note 1		R	This field shall contain the XML data.
Note 1: Determined by user.					
Type – R = REQUIRED, C = CONDITIONAL, <> surrounding the type = DESIGNATED DEFAULT VALUE ALLOWED FOR ENTIRE FIELD					

4 Appendix A

Terms and Definitions	
Term	**Definition**
Attachment Level (ALVL)	A way to associate images and graphics to the same level during movement, rotation, or display.
Base Image	A base image is the principle image of interest or focus for which other data may be inset or overlaid. The NITF file can have none, one, or multiple base images. For multiple base images in a single NITF file, the relative location of each base image is defined in the image location (ILOC) field in each image sub-header. This location will be the offset within the Common Coordinate System (CCS).
Basic Character Set (BCS)	A subset of the Extended Character Set (ECS). The most significant bit of the BCS characters is set to 0. The range of valid BCS characters code is limited to 0x20 to 0x7E plus line feed (0x0A), form feed (0x0C), and carriage return, (0x0D).
Basic Character Set-Alphanumeric (BCS-A)	A subset of the Basic Character Set. The range of allowable characters consists of space to tilde, codes 0x20 to 0x7E.
Basic Character Set-Numeric Integer (BCS-N integer)	The JBP BCS-N is a subset of the Basic Character Set (BCS) that consists of the digits '0' to '9' (codes 0x30 to 0x39), plus sign (0x2B), minus sign (0x2D), decimal point (0x2E), and slash (0x2F).
Basic Character Set – Numeric Positive Integer (BCS-N positive integer)	A subset of the BCS-N character set that consists of the digits '0' to '9' (codes 0x30 to 0x39). The most significant bit of a BCS-N character is set to 0.
BCS Space	BCS (and consequently ECS) code 0x20.
Basic Image Interchange Format (BIIF)	Format that includes the NITF2.1 specification.
Common Coordinate System (CCS)	The virtual two dimensional Cartesian-like coordinate space which will be common for determining the placement and orientation of displayable data.
Complexity Level (CLEVEL)	A code used in the file header which signals the degree of complexity an interpret implementation needs to support to adequately interpret the files. Items that differentiate complexity include: size of the common coordinate system, file size, image size, image blocking, color of imagery with greater than 8-bit per pixel, number of bands in an image segment, number of image segments, aggregate size of graphic segments, etc..
Coordinated Universal Time (UTC)	The time scale maintained by the International Earth Rotation Service (having previously been maintained by the Bureau International de l'Heure that forms the basis of a coordinated dissemination of standard frequencies and time signals.
Data Extension Segment (DES)	A type of extension segment with sub-header and data fields structured similarly to the standard data types in the NITF (e.g., image, label, symbol, text). The extension type identifier (25 character DESTAG field), the version (two character DESVER field), and the full underlying structure is under configuration management control as registered with the NTB.
ECF or ECEF	Earth Centered Fixed (ECF) or equivalently Earth Centered Earth Fixed. Coordinate system used for many SICD metadata parameters.
Extended Character Set (ECS)	A set of 1-byte encoded characters. Valid ECS character codes range from 0x20 to 0x7E, and 0xA0 to 0xFF, as well as Line Feed (0x0A), Form Feed (0x0C) and Carriage Return (0x0D). The ECS characters are described in Table B-1. As an interim measure, because of inconsistencies between standards, it is strongly advised that character codes ranging from 0xA0 to 0xFF should never be used. Therefore, the use of ECS characters should be restricted to its BCS Subset.
Extended Character Set (ECS) Space	See BCS Space definition.

Terms and Definitions	
Term	**Definition**
Extended Character Set - Alphanumeric (ECS-A)	A subset of the Extended Character Set (ECS). Valid ECS-A character codes range from 0x20 to 0x7E, and 0xA0 to 0xFF. Line Feed (0x0A), Form Feed (0x0C) and Carriage Return (0x0D) are not valid ECS-A characters. As an interim measure, because of inconsistencies between standards, it is strongly advised that character codes ranging from 0xA0 to 0xFF should never be used. Therefore, the use of ECS-A characters should be restricted to its BCS-A Subset.
Image Segment (IS)	The component of a NITF2.1 product file that contains the image pixel array or a portion of the image pixel array.
NITFS	National Imagery Transmission Format Standard
NTB	NITFS Technical Board
Non-blank	Non-blank indicates that the field cannot be filled by the character space (0x20) but may contain the character space when included with other characters.. (embedded blanks).
Null	The field is filled entirely with spaces (0x20).
Program Specific Implementation Document	A set of one of more artifacts that specify the selected metadata parameters that are included for a specific SAR sensor and/or data processing system. See NGA.STND.0024-1, Sect 1.4 for additional information.
Segment	A header and data fields.
SICD	Sensor Independent Complex Data (SICD): A format for SAR complex image data products.
SIDD	Sensor Independent Derived Data (SIDD): A format for SAR complex image data products.